

SLA²P: Self-Supervised Anomaly Detection With Adversarial Perturbation

Yizhou Wang , *Member, IEEE*, Can Qin , Rongzhe Wei , Yi Xu, *Graduate Student Member, IEEE*, Yue Bai ,
and Yun Fu , *Fellow, IEEE*

Abstract—Anomaly detection is a foundational yet difficult problem in machine learning. In this work, we propose a new and effective framework, dubbed as SLA²P, for unsupervised anomaly detection. Following the extraction of delegate embeddings from raw data, we implement random projections on the features and consider features transformed by disparate projections as being associated with separate pseudo-classes. We then train a neural network for classification on these transformed features to conduct self-supervised learning. Subsequently, we introduce adversarial disturbances to the modified attributes, and we develop anomaly scores built on the classifier’s predictive uncertainties concerning these disrupted features. Our approach is motivated by the fact that as anomalies are relatively rare and decentralized, 1) the training of the pseudo-label classifier concentrates more on acquiring the semantic knowledge of regular data instead of anomalous data; 2) the altered attributes of the normal data exhibit greater resilience to disturbances compared to those of the anomalous data. Therefore, the disrupted modified attributes of anomalies can not be well classified and correspondingly tend to attain lesser anomaly scores. The results of experiments on various benchmark datasets for images, text, and inherently tabular data demonstrate that SLA²P achieves state-of-the-art performance consistently.

Index Terms—Data mining- anomaly detection, machine learning- deep learning, representation learning.

I. INTRODUCTION

ANOMALIES, also known as outliers, are characterized as “data instances that significantly deviate from the majority of data instances” [1]. Correspondingly, anomaly detection (AD) involves identifying anomalous data points using a data-driven approach. This has been a crucial problem in machine learning for a long time and has numerous real-world applications, such

as medical health [2], [3], fraud detection [4], [5], [6], cybersecurity [7], [8] and video surveillance [9], [10], etc.

Tasks for anomaly detection is able to be broadly categorized into three groups based on the availability of labels and their degree, as follows: 1) Supervised anomaly detection involves training models on a labeled dataset which is composed of both normal and abnormal data and then utilizing them to detect anomalies in test data. 2) Semi-supervised anomaly detection (SSAD), or one-class classification, deals with the setting that the training dataset is only composed of normal data and the trained model is expected to detect outliers during the testing time. 3) Unsupervised anomaly detection (UAD), which is the most common and challenging case, complies with the condition that only unmarked data containing both typical and atypical instances are supplied, and the anomaly detection approach is required to be able to detect the outliers [11]. These three categories of tasks consider anomalies as data that belong to classes that are intrinsically distinct from the classes of the normal data. Besides, there exists another type of related but quite different detection task called out-of-distribution (OOD) detection [12], [13], [14], [15], which aims to distinguish samples that have a disjoint distribution from the training samples in the testing phase (usually from a different dataset).

We primarily concentrate on the unsupervised setting, i.e., UAD, which has the broadest applicability due to the fact that obtaining labels is expensive for anomaly detection in a many real-world scenarios [1], [16], [17]. UAD has a diverse range of applications in practice, including construction of large-scale dataset, website management, news management, etc. We must clarify that in some literature, the so-called “unsupervised anomaly detection” actually refers to SSAD (e.g., [18], [19], [20]) or OOD (e.g., [21]) by our definition, which is beyond the scope of this paper.

In this study, we present a novel SeLf-supervised outline for unsupervised Anomaly Detection employing Adversarial Perturbation, which we refer to as SLA²P. Utilizing the obtained representations from the unmarked original data, we project them into distinct subspaces through multiplying random matrices. Despite having no explicit idea of the subspaces, we deem the transformation to be digging various unknown aspects of latent information of the data. Subsequently, we train a deep neural network (DNN) classifier upon the transformed features to identify which specific subspace they are projected into. The training procedure of the classifier network is equipped with an early stopping technique to refrain from overfitting, and

Received 24 April 2023; revised 30 May 2024; accepted 6 August 2024. Date of publication 23 August 2024; date of current version 13 November 2024. This work was supported by the Air Force Office of Scientific Research under Grant FA9550-23-1-0290. Recommended for acceptance by M.A. Cheema. (Corresponding author: Yizhou Wang.)

Yizhou Wang, Can Qin, Yi Xu, and Yue Bai are with the Department of Electrical and Computer Engineering, Northeastern University, Boston, MA 02115 USA (e-mail: wyzjack990122@gmail.com; qin.ca@northeastern.edu; xu.yi@northeastern.edu; bai.yue@northeastern.edu).

Rongzhe Wei is with the Georgia Institute of Technology, Atlanta, GA 30332 USA (e-mail: rwei42@gatech.edu).

Yun Fu is with the Department of Electrical and Computer Engineering, Khoury College of Computer Science, Northeastern University, Boston, MA 02115 USA (e-mail: yunfu@ece.neu.edu).

The code is available at: <https://github.com/wyzjack/SLA2P>.

This article has supplementary downloadable material available at <https://doi.org/10.1109/TKDE.2024.3448473>, provided by the authors.

Digital Object Identifier 10.1109/TKDE.2024.3448473

the reason behind this is that outliers fail to be trained well owing to their smaller population size and more diverse modes. The trained classifier is expected to primarily acquire beneficial information concerning characteristics of the normal data and consistently make inaccurate predictions on anomaly data. For normal data, the early stopping mechanism, with a threshold between 0 and 1, allows for a mix of accurate and inaccurate predictions due to the higher volume of normal samples. Hence the predictive uncertainties can be employed to create anomaly scores, where anomalies exhibit more consistent predictions, while normal instances display more distinct predictions. The differences in prediction distributions can be intensified by incorporating adversarial perturbations to the transformed features. The aim of the perturbations is to decrease the softmax values corresponding to the anticipated labels of the transformed features by the classifier, which are empirically shown to boost detection performance. To summarize, our contributions are three-fold:

- We introduce an innovative framework SLA²P for UAD, which can be applied to the image, text, and inherently tabular datasets. Our approach is the first attempt to use **feature-level transformation** to create a pretext classification task as surrogate supervision for the Unsupervised Anomaly Detection task.
- We investigate UAD task by involving **adversarial perturbation** in a **self-supervised** fashion to achieve considerable performance improvement. We are not aware of any other analogous approach in the anomaly detection field.
- Our approach attains cutting-edge results on 7 challenging datasets, consistently surpassing the current best methods by a considerable margin. Besides, it is **highly robust** and can maintain excellent detection performance under particularly high or small anomaly ratios.

The prior edition of this paper is presented in [22]. The main extensions include: (1) the theoretical foundation of our proposed SLA²P approach (Section I) on the similarity-preserving property of random projections which connects it with geometric transformations, (2) new experimental results when the normal data are multimodal, which further verifies that our approach can work beyond unimodal setting. (3) freshly designed ablation study about the robustness of the performance of SLA²P w.r.t. the transformed dimension which echos Theorem 1, (4) a comprehensive related work demonstrated in terms of unsupervised anomaly detection, anomaly detection with pretrained networks, self-supervised learning and input perturbation.

II. RELATED WORKS

Unsupervised anomaly detection Conventional approaches to UAD are based mostly on classic unsupervised learning tools, encompassing density estimation techniques [23], [24], [25], clustering methods [26], [27], dimensionality reduction methods [28], [29] and one-class SVM methods [30], [31]. Owing to the extraordinary expressive capacity of DNNs, a lot of reconstruction-based techniques that use DNNs have been created for UAD recently. They mainly employ deep generative models [32], [33], [34] or autoencoders (AE) [35], [36], [37]

to reconstruct data and determine abnormality of data via its reconstruction error. For instance, DAGMM [38] feed the AE latent representations into a GMM and cooperatively optimize them. RSRAE [39] incorporates a robust Principal Component Analysis layer [40] into deep autoencoders, which is intended to project the inliers into their subspace while leaving the outliers out. However, it is unavoidable that such kind of reconstruction-based approaches concentrate more on element-wise or low-level error instead of semantic information in the high level, as pointed out by [41].

Anomaly detection with pretrained networks Transferring discriminative embeddings of pretrained nets to AD has been widely studied and achieved great success [42], [43], [44]. Andrews et al. [45] shows that transfer-representation-learning approaches offer viable representations for AD tasks without prior knowledge of the data. Methods in [46], [47] improve AD performance on target domains via transferring information of related domains. Burlina et al. [48] employs pretrained VGGNets [49] to perform novelty detection. Recently DNNs pretrained on ImageNet have been used to extract features for unsupervised anomaly detection and segmentation on images [20], [50], [51], [52] and videos [53], [54].

Self-supervised learning (SSL) has been an increasingly prevailing unsupervised learning method. SSL methods learn representations via designing a pretext task between inputs and self-defined signals [55], [56] or contrastive learning [57], [58]. Employing SSL techniques to assist semi-supervised AD has shown promising results recently [59], [60], [61], [62]. In UAD task, E³Outlier [41] trains a Convolutional Net via SSL and uses the network outputs of the CNN to design anomaly scores. Nevertheless, such practice can only be applied to image data and requires manually pre-defined transformations. By contrast, our method can be applied to both image and tabular data, and in our framework, random transformations are able to exhibit promising performance.

Input Perturbation: Input adversarial perturbation technique is proposed by [63] for the first time to yield adversarial data samples to dupe the classifier. In OOD detection literature, several works take advantage of the opposite perturbation to the raw input data employing the gradient of the maximum softmax score of the label predicted from pre-trained network [12], [14]. Lee et al. [64] add perturbation to increase the proposed Mahalanobis distance-based confidence score for OOD detection. In the anomaly/outlier detection area, there are not many efforts to use input-level perturbations to enhance AD performance as we fail to directly use some pre-trained network softmax outputs. As far as we know, we are the first to seamlessly combine the perturbation technique with self-supervised learning to address anomaly detection problems.

III. PROBLEM STATEMENT

We contemplate data space $\mathcal{X} \subset \mathbb{R}^d$ and we are provided an dataset $X \subset \mathcal{X}$ which is unlabeled. X is composed of both inliers (normal data) X_{in} of size n and outliers (anomalous data) X_{out} of size np , where $0 < p < 1$ is the anomaly ratio which is the ratio of the number of the outliers to that of the inliers. The

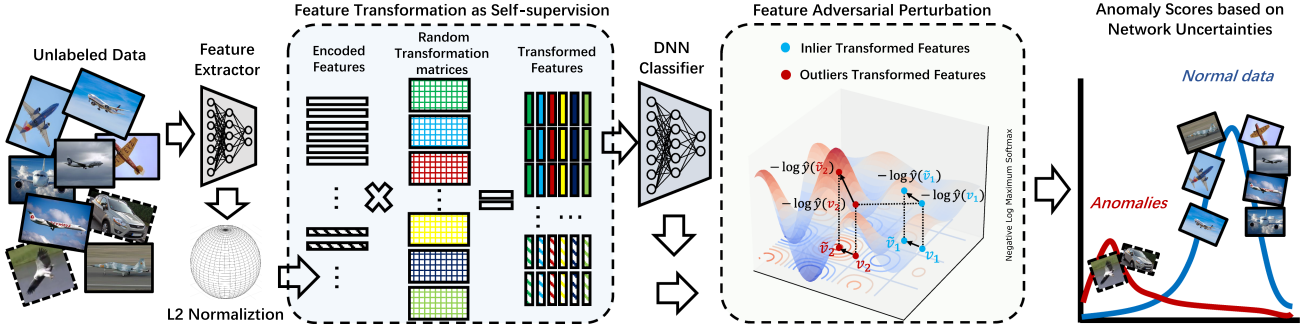


Fig. 1. Overview of our proposed framework SLA^2P on image data. Given unlabeled image data with both inliers and outliers, we first extract representative features from raw data and normalize them into a unit sphere vector space (features of anomalies are shaded). We then apply random projections to the features through multiplying matrices randomly sampled from standard normal distribution. Transformations by different matrices result in pseudo labels, on which we train a DNN classifier. Next, we adversarially perturb the transformed samples using the gradients of the softmax output scores of the predicted labels of the classifier. Finally, the scores for identifying anomalies are produced by utilizing the uncertainty estimates of the network's predictions on the altered transformed characteristics.

primary goal of UAD is to create an indicator function $I(\mathbf{x})$ that is universally applicable: $\mathcal{X} \rightarrow \{0, 1\}$ such that $I(\mathbf{x}) = 1$ for $\mathbf{x} \in X_{in}$ and $I(\mathbf{x}) = 0$ for $\mathbf{x} \in X_{out}$. However, directly pursuing such function is hard and inefficient, as there is a compromise existing between type-I error (rate of normal samples which are classified as anomalies) and type-II error (rate of anomalous samples that are classified as normal) [59]. The standard practice to deal with this problem is to instead design a score function $S(\mathbf{x}) : \mathcal{X} \rightarrow \mathbb{R}$ such that lower scores indicate more abnormality while higher scores indicate more normality.

IV. METHOD

A. Utilizing L2 Normalization for Feature Extraction

Fig. 1 illustrates our proposed outline SLA^2P . Provided a raw datum $\mathbf{x}$, we transform it into its distinguishing representation as an embedding $\mathbf{f}(\mathbf{x}) \in \mathbb{R}^d$ using a feature extractor $\mathbf{f}$. The detailed choice setting of the embedding extraction function f for different datasets is elaborated in Section VI. L2 normalization is implemented on the features after embedding extraction:

$$\mathbf{v}(\mathbf{x}) = \frac{\mathbf{f}(\mathbf{x})}{\|\mathbf{f}(\mathbf{x})\|_2}.$$

The embeddings are projected onto a sphere in $\mathbb{R}^d$ through L2 normalization, which merely decreases the dimension of the feature by one but gets rid of the magnitude discrepancies among the unlabeled data [65]. Our idea is that anomalies differ from normal data in intrinsic and latent distribution, not magnitudes. Therefore the normalization step is supposed to help the subsequent modules of SLA^2P to concentrate on semantic level discrepancies and attenuate low-level variances between normal and anomalous data. Our theoretical argument also corroborates its effectiveness in that the proof of our similarity preserving Theorem 1 requires the L2-norm of the processed vectors to be 1 (see Section V for more details).

The function $f(\mathbf{x})$ can be any embedding function that converts raw input data into vector representations with semantic information. The selection of models or approaches for feature extraction is not constrained, as our SLA^2P framework is designed to operate effectively regardless of the feature extractor

employed. For inherently tabular data, the raw data is qualified to serve as its embedding, i.e., choosing $\mathbf{f}$ to be identity mapping is enough.

B. Self-Supervised Learning At the Feature Level

Inspiration drawn from geometric transformations in images: Geometric transformations as self-supervision have obtained tremendous success in unsupervised learning of image data [41], [56], [59] and the central concept revolves around the notion that DNNs are thought to have the ability to extract valuable information from image data if they can detect their transformed patterns [56]. Nevertheless, the majority of data utilized in machine learning tasks is presented in a tabular format that does not have a structure identical to that of two-dimensional images, and there exist no in-hand “geometric transformations” for vector-form data. In fact, geometric transformations are executed via implementing affine transformations on the homogeneous coordinates of images. Given a coordinate pair $(x, y)^T$, it is first symbolized as a three-dimensional vector through adding an additional dimension with element 1. Then the new coordinates $(x', y')^T$ are generated through the multiplication of a transformation matrix:

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

When $a = \cos \theta, b = \sin \theta, d = -\sin \theta, e = \cos \theta, c = f = 0$, the manipulation carries out rotation by angle θ ; when $a = e = 1, b = d = 0, c = x_0, f = y_0$, it conducts image translation along the x axis by x_0 and along the y axis by y_0 .

Motivated by this point, we suggest an extension of this type of transformation to data in vector form by substituting the coordinate vectors with the vectors themselves. Consequently, we are able to execute transformations from the feature level to perform self-supervised learning. The designing of the transformation matrix still remains a problem since a rotation matrix of vectors does not have the same meanings as a rotation matrix for image 2D coordinates. Intriguingly, our experiments show that employing random matrices of which each element is independent and

identically distributed (i.i.d.) sampled from the standard normal distribution, is sufficient to produce exceptional results. Next, we give a detailed description of using random matrices for self-supervision.

Random Matrix Multiplications as Surrogate Supervision: We denote the random matrix set as $\mathcal{A} = \{\mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \dots, \mathbf{A}^{(M)}\}$. For any $1 \leq m \leq M$, $\mathbf{A}^{(m)} = (a_{ij}) \in \mathbb{R}^{k \times d}$ ($k \ll d$) are i.i.d. sampled and the components of the matrix are also sampled i.i.d. from standard normal distribution

$$a_{ij} \sim \mathcal{N}(0, 1).$$

The random transformation we create is intended to be the operation of multiplying the aforementioned random matrices to data features:

$$\mathbf{v}^{(m)} = \mathbf{A}^{(m)} \mathbf{v},$$

As a result, we have created a self-labeled dataset

$$D_{\mathcal{A}} \triangleq \left\{ \left(\mathbf{v}^{(m)}(\mathbf{x}), m \right) \mid \mathbf{x} \in \mathcal{X}, \mathbf{A}^{(m)} \in \mathcal{A} \right\}.$$

Here we treat m as the pseudo label of $\mathbf{A}^{(m)} \mathbf{v}$, and therefore We possess a self-determined classification dataset featuring $Mn(1+p)$ transformed features. The random projections serve a dual purpose of projecting feature embeddings into distinct spaces and reducing their dimensions. We observe that a random matrix multiplication projects one subspace or convex set in $\mathbb{R}^d$ to another subspace or convex set in $\mathbb{R}^k$, as claimed in Proposition 1. Considering that extracted features of normal data tend to be in a certain subspace or convex set, Proposition 1 guarantees that the transformed features of normal data are still in one subspace or convex set.

Proposition 1. For any matrix $\mathbf{A} \in \mathbb{R}^{k \times d}$, linear subspace $\mathcal{V} \subset \mathbb{R}^d$ and convex set $\mathcal{U} \subset \mathbb{R}^d$, $\mathbf{A}\mathcal{V}$ is a linear subspace in $\mathbb{R}^k$ and $\mathbf{A}\mathcal{U}$ is a convex set in $\mathbb{R}^k$.

Pseudo label classifier training We can straightforwardly train a multi-class classifier C_{θ} with parameters θ on the dataset which is self-labeled, i.e., the classifier network is supposed to categorize the transformed features $\{\mathbf{v}^{(m)}(\mathbf{x}_i)\}$ into pseudo-class m for any $1 \leq m \leq M$. We adopt the cross-entropy loss as the loss function. The output vector generated by the softmax function of C_{θ} is denoted as $\mathbf{y}(\cdot|\theta)$ with m th element $y^{(m)}(\cdot|\theta)$, and then the pretext task we define for ourselves is able to be formulated as

$$\min_{\theta} \frac{1}{n(1+p)} \sum_{i=1}^{n(1+p)} \mathcal{L}(\mathbf{x}_i|\theta),$$

where

$$\begin{aligned} \mathcal{L}(\mathbf{x}_i|\theta) &= -\frac{1}{M} \sum_{m=1}^M \log \left(y^{(m)} \left(\mathbf{v}^{(m)}(\mathbf{x}_i) \mid \theta \right) \right) \\ &= -\frac{1}{M} \sum_{m=1}^M \log \left(y^{(m)} \left(\mathbf{A}^{(m)} \mathbf{v}(\mathbf{x}_i) \mid \theta \right) \right). \end{aligned}$$

In the training stage, we use an early stopping strategy to abstain from the overfitting problem. We set a hyperparameter

μ as a threshold of classification accuracy so that the training process terminates once the categorization precision of the classifier network of the current batch gets to $\mu \in (0, 1)$. This strategy actually holds a crucial function within our framework because if the classifier C_{θ} is capable of discerning all the transformed data well, the classification outcome struggles to effortlessly identify the irregular data.

The random projections transform the extracted features of the data into various subspaces, and the DNN classifier is subsequently trained to discern the specific subspace into which these features have been projected. This methodology draws inspiration from the 2D geometric transformations prevalent in computer vision literature, which encompasses numerous studies focusing on applying geometric transformations to images and training CNNs to recognize the specific transformations applied for representation learning [56], [59], [66]. The underlying rationale for these methods is that if a CNN can accurately predict the applied geometric transformation, the network has acquired robust orientation-invariant features of the image. Analogously, random projections can be perceived as geometric transformations of vector representations within vector space. Therefore, if our DNN classifier can successfully identify the applied transformations, it is indicative of the network's ability to learn robust rotation-invariant features of the vector representations.

C. Adversarial Perturbation-Driven Scoring Using Network Uncertainty

Our anomaly score function $S(\mathbf{x})$ is defined based on the predictive uncertainties of the classifier C_{θ} . Our goal is to assign higher scores to normal data compared to anomalies. Rather than employing the outcomes from training on the initial transformed examples like [41], [59] did, We incorporate adversarial perturbation into the modified attributes prior to inputting them into the classification model C_{θ} .

Adversarial Perturbation to the Transformed Features: We adversarially perturb the transformed features using the gradient of the negative log softmax score of the predicted class of the trained classifier C_{θ} w.r.t. the input sample. In mathematical terms, for any $1 \leq m \leq M$ and $\mathbf{x}$, we let

$$\tilde{\mathbf{v}}^m(\mathbf{x}) = \mathbf{v}^{(m)}(\mathbf{x}) + \eta(-\nabla_{\mathbf{v}^{(m)}(\mathbf{x})} \log \hat{y}^{(m)}(\mathbf{x}|\theta)), \quad (1)$$

where $\hat{y}^{(m)}(\mathbf{x}|\theta) = \max_i y^{(i)}(\mathbf{v}^{(m)}(\mathbf{x})|\theta)$, and η is the perturbation magnitude. Considering that C_{θ} has been trained comparatively well on the majority of the transformed data samples, this approach intends to reduce the pseudo-class's softmax score that has the highest predicted probability, which means to increase the difficulty in classifying the projected features. Note that our adversarial perturbation form is different from [63] as we do not involve the pseudo labels when computing the gradient. Empirically, we discover that following this operation, the anomaly scores for both outliers and inliers will experience a reduction, however, inliers will exhibit greater resilience to perturbations than outliers.

Negative Brier Score: Stimulated by reconstruction-based anomaly detection methods [35], [38], [39] which use L2 distance error of input and reconstruction as score, we employ euclidean distance for anomaly scoring by means of viewing the one-hot embedding of the correct label as ground truth within the classification task:

$$S(\mathbf{x}) = -\frac{1}{M} \sum_{m=1}^M \left\| \mathbf{y} \left(\tilde{\mathbf{v}}^{(m)}(\mathbf{x}) | \boldsymbol{\theta} \right) - \mathbf{e}_m \right\|_2^2,$$

where $\mathbf{e}_m$ is the m -th canonical basis vector representing the label vector of m -th pseudo-class. $-S(\mathbf{x})$ refers to the mean squared error of the output predictions of C_θ and one-hot ground truth pseudo labels, corresponding to a classic proper score rule known as Brier Score [67]. The ideal case is that the normal data have lower errors and, accordingly, higher anomaly scores.

V. THEORETICAL FOUNDATION OF SLA²P

Random projections do not manifestly preserve semantic information of data, which contrasts with geometric transformations (e.g., a rotated or translated plane is still a plane, not a bird). Due to randomness, one may have no explicit idea of what subspace the features are projected into and whether the random transformations have an influence on the discrepancies among original features. Intriguingly, we find that as long as the dimension k is sufficiently large, the L2 distance and inner product of transformed embeddings, which are two effective measures of similarities between high-dimensional vectors, are roughly in proportion to their original values by a constant factor k .

Theorem 1 (Similarity-preserving Property): Given any fixed m and index pair of the unlabeled dataset (i, j) , for any $0 < \epsilon < 1$ and $0 < \delta < 1$,

1) when $k > \frac{4 \log \frac{2}{\delta}}{\epsilon^2 - \epsilon^3}$, we have with at least $1 - \delta$ probability over the random sampling of matrix $\mathbf{A}^{(m)}$,

$$(1 - \epsilon) \|\mathbf{v}_i - \mathbf{v}_j\|_2^2 \leq \frac{\|\mathbf{v}_i^{(m)} - \mathbf{v}_j^{(m)}\|_2^2}{k} \leq (1 + \epsilon) \|\mathbf{v}_i - \mathbf{v}_j\|_2^2,$$

2) when $k > \frac{4 \log \frac{4}{\delta}}{\epsilon^2 - \epsilon^3}$, we have with at least $1 - \delta$ probability over the random sampling of matrix $\mathbf{A}^{(m)}$,

$$\mathbf{v}_i \cdot \mathbf{v}_j - \epsilon \leq \frac{\mathbf{v}_i^{(m)} \cdot \mathbf{v}_j^{(m)}}{k} \leq \mathbf{v}_i \cdot \mathbf{v}_j + \epsilon.$$

The supplementary contains the proof of Theorem 1, which is based on Johnson–Lindenstrauss lemma [68]. As stated in Theorem 1, random projections are likely to preserve the similarities among features, i.e., originally distant features (e.g., one normal and one anomalous instance) will still be comparatively distant, and features originally closely distributed (e.g., normal data) tend to be near each other after projection. This lays the theoretical foundation of our SLA²P framework in that, although the transformations are random, they still preserve the structure and inner relationship of the data, which is coherent with hand-crafted 2D geometric transformations. We leave the theoretical analysis on probabilities/error of our method for future work.

VI. EXPERIMENTS

We conduct diverse experiments to illustrate the effectiveness and generality of our SLA²P approach for unsupervised anomaly detection.

A. Datasets

We conducted an empirical evaluation of our framework, SLA²P, on three image datasets and four tabular datasets to demonstrate its capability and generality. The datasets of image data are CIFAR-10 [69], CIFAR-100 [69] and Caltech 101 [70]. The datasets of tabular data consists of text datasets 20 Newsgroups [71], Reuters-21578¹ and inherently tabular datasets KDDCUP99 [72] and Arrhythmia. For the text and image classification datasets, each category of data serves as inliers in turn, and the reported outcome is the mean performance across all classes. The detailed descriptions are as follows.

CIFAR-10 is composed of 60,000 images distributed across 10 categories with resolution 32×32 . There are 6,000 images per class. In each experiment, the inliers are 6,000 images from one class, and the outliers are 6000*p* images randomly selected from the rest classes.

CIFAR-100 contains 60,000 images of resolution 32×32 labeled according to 100 distinct categories. The 100 categories are grouped into 20 superclasses and there are 3,000 images within each superclass. In each experiment, the inliers are 3,000 images from one superclass, and the outliers are 3000*p* images randomly chosen from the rest of the superclasses.

Caltech 101 contains 9,146 images of 101 different classes. Each image's size is 300×200 . Following the experiment setting of [39], we select 11 classes that consist of at least 100 images and randomly sample 100 out of them for each class. In each individual experiment, the inliers are 100 images from one certain class, and the outliers are $100 \times p$ images selected from the rest ten classes at random.

20 Newsgroups is a text classification dataset with approximately 20,000 evenly distributed newsgroup articles, divided across 20 classes. We randomly sample 360 documents per class. In each individual experiment, the inliers are 360 documents from one class, and the outliers are 360*p* documents randomly chosen from the rest of the classes.

Reuters-21578 is a text classification dataset containing 90 text categories. We choose the five largest single-label classes and randomly sample 360 documents for each class. In each individual experiment, the inliers data are the documents from one class, and 360*p* outliers are sampled randomly from the remaining classes.

Arrhythmia is a small-scale medical dataset containing attributes on the diagnosis of cardiac arrhythmia in patients and includes 16 classes. We construct the anomalous dataset using the smallest classes 3,4,5,7,8,9,14,15 and the normal set using the rest. There are 452 data instances among which 66 are anomalous.

¹<http://www.daviddlewis.com/resources/testcollections/reuters21578/>

KDDCUP99 is a large-scale intrusion detection dataset. Following [38] we utilize the full UCI 10% dataset, where the non-attack classes are treated as anomalies. There are 97278 abnormal instances and 396743 normal ones.

B. Experiment Setup

We contrast SLA²P against six state-of-the-art techniques for anomaly detection: IF [75], OCSVM [30], [31], DAGMM [38], E³Outlier [41], RSRAE [39] and PANDA [76]. We adopt the commonly used Area under the Receiver Operating Characteristic curve (AUROC) and Area under the Precision-Recall curve (AUPR) as the assessment criteria. For AUPR, we consider outliers as “positive” when computing. All the experiments are **conducted independently 5 times using the same random seeds 0, 1, 2, 3, 4 for fair comparison**, and the averaged scores are reported in the main paper. More experimental details (including each single classwise experiment result) are provided in the supplementary file.

Hyperparameter Tuning Rule: We implement a K -fold cross-validation strategy for hyperparameter tuning. This method involves dividing the dataset into l subsets. For each fold, $K - 1$ subsets are used for training, while the remaining subset is designated as the validation set. This process is repeated K times, ensuring each subset is used as the validation set exactly once. The consistency of anomaly scores across the different subsets of the dataset serves as our internal validation metric. Specifically, we calculate the variance of the variance of the anomaly scores for each fold, selecting the hyperparameter set that minimizes this metric. For all the individual experiments, the hyperparameter search range is defined as follows: for M , the candidate set is $\{64, 128, 256, 512\}$; for k , it is $\{128, 256\}$; for μ , it is $\{0.3, 0.4, 0.5, 0.6, 0.75, 0.9\}$; and for η , it is $\{10, 100, 1000, 10000\}$. We choose the fold number $K = 5$ for all the experiments. The motivation of our hyperparameter tuning rule is that the hyperparameter configuration exhibiting the minimum variance of variance across multiple folds ensures that our model’s performance remains stable and consistent across various data splits.

Setup of SLA²P: For image data, we use ResNets [77] pre-trained using ImageNet dataset without the last fully connected layer to obtain embeddings. To match the image input shapes, we first resize raw images to 224×224 using Bilinear Interpolation and then feed it into the extractor network. We preprocess raw text data into vectors by sequentially utilizing the TFIDF transformer and Hashing-vector techniques [78], which is the same process as in [39]. When dealing with data that is inherently in a tabular format, we use the raw data vectors as input without any further processing. We also conduct experiments using modern pretrained language model for embedding extraction. We take BERT [73] and GPT-3 [74] which are the two main-stream and archetypal pretrained LLMs. Specifically, for BERT, we use “bert-base-uncased” model from package transformers;² for GPT-3, we use “text-embedding-ada-002” model from package openai.³ For Reuters dataset, we choose $M = 512$ and $k = 128$

because of high dimensionality. For KDDCup99 dataset, we assign $M = 64$ and $k = 128$ due to high computational burden. For the remaining datasets, we set $M = 256$ and $k = 256$. As to the early stopping threshold, we set $\mu = 0.75$ on 20 Newsgroups dataset, $\mu = 0.3$ on Reuters dataset, and $\mu = 0.6$ on all the remaining datasets. We select the perturbation magnitude as $\eta = 10$ for 20 Newsgroups, $\eta = 1e2$ for Reuters, $\eta = 1e4$ for CIFAR-100 and $\eta = 1e3$ for all the remaining datasets. In fact, setting $\mu = 0.6$ and $\eta = 1e3$ is universally adequate for different UAD tasks, and there is an implicit relationship between these two hyperparameters (see ablation study). Given that the input dimension of the classifier network is q , the classifier network used in all experiments is a three-layer fully connected network, which has the following structure: FC($q, 2q$)-FC($2q, 4q$)-FC($4q, M$). Batch normalization [79] is applied to each layer and LeakyReLU [80] is employed as activation function. We use Adam [81] for the training with step size $1e-3$ and weight decay parameter $5e-4$.

Setup of Baseline Methods: Our experiments indicate that all the baseline methods, except E³ outlier (which directly manipulates images), work better using extracted representative embeddings as input than using raw data as input. Hence to enable a fair comparison, we evaluate our method against the baseline approaches with extracted features as input in the sequel. For the implementation of the benchmarks, we adapt the code from package scikit-learn [82] for IF and OCSVM. We implemented DAGMM and E³Outlier using the code⁴ from [41] with minimal modifications such that they are applicable to our datasets and adapt to our experimental protocol. RSRAE results are obtained running the official code.⁵ We adapt PANDA for UAD setting by setting the testing set and the training set to be the same with both inliers and outliers using the official code.⁶ All the data pre-processing processes follow the original papers.

C. Results

The average AUROC and AUPR of all the approaches on all the datasets are summarized by us. To manifest the broad applicability of our method, we show its performance under different anomaly ratios $p = 10\%, 30\%, 50\%$. We call our method without the adversarial perturbation step as SLA and include its performance as well. Results on image datasets using pre-trained ResNet-50 network are given in Table I. The results using pre-trained ResNet-101 are provided in the supplementary for completeness, where slightly better performances are achieved owing to the usage of a deeper pre-trained DNN extractor. We also summarize the results on text datasets and inherently tabular datasets in Tables II, III, IV, and V respectively.

Results on Image Datasets: As shown in Table I, the performance of our two variants of our proposed approaches, especially SLA²P, consistently exceed existing approaches on all the image datasets, including CIFAR-10 and CIFAR-100, which are abidingly considered the most challenging datasets for UAD [41]. In addition, SLA²P is robust to the anomaly rate and

²<https://pypi.org/project/transformers/>

³<https://pypi.org/project/openai/>

⁴<https://github.com/demonzyj56/E3Outlier>

⁵<https://github.com/dmzou/RSRAE>

⁶<https://github.com/talreiss/PANDA>

TABLE I
AUROC/AUPR (%) PERFORMANCE ON IMAGE DATASETS FOR UAD WITH RESNET-50 AS THE FEATURE EXTRACTOR

Dataset	p	IF	OCSVM	DAGMM	E ³ Outlier	RSRAE	PANDA	SLA (ours)	SLA ² P (ours)
CIFAR-10	10%	83.24/43.55	85.19/47.57	59.86/15.73	85.89/45.09	83.58/42.48	86.69/59.43	<u>90.81/59.46</u>	92.21/68.22
	30%	79.36/56.95	79.52/56.69	67.04/37.81	80.58/58.53	75.05/50.90	83.63/64.20	<u>87.68/70.08</u>	89.11/72.96
	50%	75.94/62.25	74.83/60.04	62.73/44.84	77.44/64.26	71.27/56.92	82.75/68.20	<u>84.64/73.25</u>	86.63/76.95
CIFAR-100	10%	77.35/33.35	79.98/35.58	54.47/12.09	80.36/33.61	89.21/55.94	84.70/58.33	<u>87.51/52.13</u>	91.45/71.70
	30%	73.90/49.65	74.86/49.62	56.78/29.59	78.60/53.15	85.33/66.40	82.32/64.21	85.03/66.39	85.48/69.29
	50%	70.81/56.51	71.21/55.96	55.13/38.16	76.22/61.00	81.68/69.46	77.83/68.88	82.48/71.21	<u>81.98/70.76</u>
Caltech 101	10%	89.88/59.72	92.98/67.16	72.61/34.02	90.72/62.29	94.28/74.44	96.32/80.39	<u>96.36/80.98</u>	96.42/81.12
	30%	87.12/69.86	86.80/69.40	73.53/49.20	87.37/70.97	87.52/69.94	94.67/85.72	<u>95.09/86.17</u>	95.35/87.70
	50%	84.29/72.97	81.67/69.82	69.14/53.70	84.82/73.93	82.59/71.78	<u>93.97/88.91</u>	93.57/87.73	96.26/94.31

The best results are in bold and the second best ones are underlined. The average scores across five separate trials are presented.

TABLE II
AUROC/AUPR (%) PERFORMANCE ON TEXT DATASETS FOR UAD USING CLASSIC EMBEDDINGS

Dataset	p	IF	OCSVM	DAGMM	RSRAE	SLA (ours)	SLA ² P (ours)
20 Newsgroups	10%	56.45/15.09	82.04/43.82	54.84/12.76	89.03/49.79	<u>90.77/50.57</u>	93.20/71.77
	30%	58.35/34.56	75.07/52.28	54.07/27.38	88.86/68.61	86.29/63.58	90.16/74.21
	50%	57.00/43.77	71.33/58.49	51.94/35.93	86.73/73.55	84.48/71.18	<u>86.55/78.13</u>
Reuters-21578	10%	59.30/18.74	90.95/61.69	65.07/22.33	92.44/64.44	<u>93.51/64.58</u>	95.77/81.40
	30%	58.86/35.99	83.24/63.25	59.33/33.61	<u>88.70/73.07</u>	88.07/71.95	95.52/89.68
	50%	55.85/42.10	74.88/62.36	62.01/46.54	<u>83.39/72.93</u>	82.87/72.64	86.09/79.92

TABLE III
AUROC/AUPR (%) PERFORMANCE ON 20 NEWSGROUPS DATASET FOR UAD USING PRETRAINED LM EMBEDDINGS

	p	IF	OCSVM	DAGMM	RSRAE	SLA (ours)	SLA ² P (ours)
BERT [73]	10%	55.76/10.93	59.62/16.47	54.37/10.50	60.53/12.47	<u>66.89/14.57</u>	69.51/19.90
	30%	53.85/18.44	56.91/18.14	52.22/17.92	52.29/23.78	<u>57.21/24.80</u>	60.03/29.43
	50%	52.26/26.84	54.63/29.42	50.98/28.88	51.50/33.27	<u>55.64/32.03</u>	59.50/41.45
GPT-3 [74]	10%	58.82/16.96	83.58/42.54	56.11/15.42	88.25/47.82	<u>90.62/66.96</u>	92.90/69.03
	30%	59.72/36.24	78.29/54.68	55.60/28.57	87.96/69.72	<u>91.71/69.94</u>	94.89/74.65
	50%	57.43/45.11	72.92/59.27	53.93/33.70	86.24/74.43	<u>86.58/74.00</u>	87.62/77.84

TABLE IV
UAD PERFORMANCE ON INHERENTLY TABULAR DATASETS

	Arrhythmia		KDDCUP99	
	AUROC (%)	AUPR (%)	AUROC (%)	AUPR (%)
IF	79.78±0.93	46.29±1.30	92.72±1.17	69.06±3.91
OCSVM	79.47±0.00	<u>47.75±0.00</u>	51.98±0.00	43.23±0.00
DAGMM	67.16±4.14	30.68±1.98	77.55±5.90	47.28±5.80
RSRAE	<u>80.31±2.89</u>	42.03±2.84	60.09±2.30	27.36±1.22
SLA (ours)	79.85±0.69	46.78±1.22	97.00±0.46	<u>88.89±0.92</u>
SLA ² P (ours)	98.93±0.08	90.59±0.66	97.59±0.34	90.59±0.86

TABLE V
UAD PERFORMANCE ON ARRHYTHMIA USING PRETRAINED LM EMBEDDINGS

	BERT [73]		GPT-3 [74]	
	AUROC (%)	AUPR (%)	AUROC (%)	AUPR (%)
IF	79.73±0.90	42.74±0.60	81.46±0.25	44.19±0.42
OCSVM	80.89±0.00	48.30±0.00	83.04±0.00	49.57±0.00
DAGMM	72.34±0.52	34.75±0.69	76.34±0.96	38.43±0.70
RSRAE	78.50±1.36	43.24±2.01	79.88±0.58	51.50±0.91
SLA (ours)	<u>88.11±0.38</u>	<u>64.38±0.59</u>	<u>84.58±0.14</u>	<u>56.72±0.94</u>
SLA ² P (ours)	99.88±0.01	99.07±0.14	99.75±0.04	97.41±0.67

can still obtain outstanding performance when the anomaly ratio p is large, with AUROC score 86.63% on CIFAR-10, 82.48% on CIFAR-100 dataset and 96.26% on Caltech 101 dataset when the anomaly count reaches half the number of normal samples. This

is particularly commendable as when the number of anomalies increases, no matter what UAD method is used, it is inevitable that the adverse effect of the learning process of outliers to the training of inliers will increase accordingly [34], [39].

Results on Tabular Datasets: Our framework demonstrates superior performance not only on image datasets but also on tabular datasets, as evidenced by Tables II, III, IV, and V. Specifically, SLA²P attains state-of-the-art (SOTA) outcomes on 20 Newsgroups whether classic embedding or pretrained language model embedding is used. When using classic embedding, SLA²P surpasses the existing SOTA RSRAE by 3%-4% in AUROC and approximately 10% in AUPR for varying anomaly proportions. On Reuters, Our approach demonstrates a notable improvement over the existing SOTA, achieving as much as 6.8% increase in AUROC and 16.6% enhancement in AUPR when $p = 0.3$ compared to RSRAE. When using pretrained LM embeddings, SLA²P achieves SOTA results in both AUROC and AUPR consistently on the 20 Newsgroups dataset, which indicates its generality. Moreover, our method exhibits dominating performance on inherently tabular datasets, with more than 40% AUPR improvement on Arrhythmia and in excess of 20% AUPR increase on KDDCUP99 when raw data serves as input. The AUROCs and AUPRs of these two classical datasets all exceed 90%, showcasing the practicality of SLA²P on large-scale datasets such as KDDCUP99. When using pretrained LM embeddings,

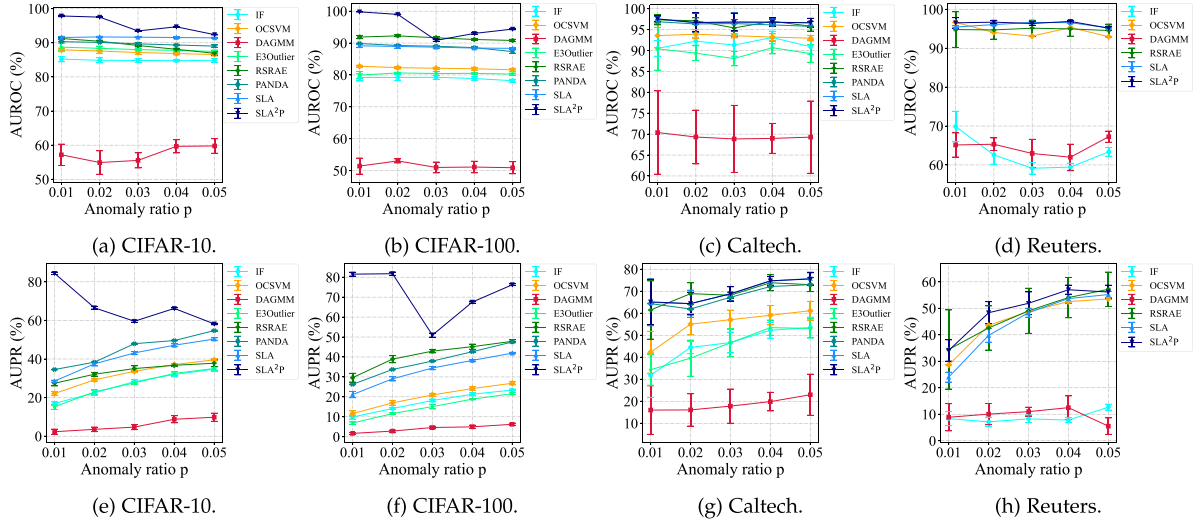


Fig. 2. AUROC and AUPR when the anomaly ratio is tiny.

TABLE VI
AUROC/AUPR (%) WHEN THE INLIERS OBEY MULTIMODAL DISTRIBUTION

Dataset	IF	OCSVM	DAGMM	RSRAE	SLA (ours)	SLA ² P (ours)
Caltech	38.10/21.12	40.46/20.93	52.38/34.92	57.80/35.74	<u>61.06/43.35</u>	71.50/65.89
20News	48.43/23.09	38.16/18.55	52.75/30.71	60.86/38.97	<u>65.05/44.77</u>	84.12/54.70
Reuters	50.87/25.86	47.99/24.51	53.71/30.95	52.49/27.63	<u>67.05/46.57</u>	86.86/61.38

SLA²P demonstrates dominant performance whether BERT or GPT-3 is used, exceeding all the current SOTA approaches. It is worth noticing that, pretrained embedding might not always perform better in our task than classic embedding. For instance, on the 20 Newsgroups dataset, BERT embeddings lead to significantly worse results than classic embeddings generated utilizing TFIDF transformer and Hashing-vector techniques.

Results Under Tiny Anomaly Ratios: The proportion of anomalies may be exceptionally low in some unsupervised anomaly detection tasks and circumstances. Therefore we further compare our methods to the baseline approaches when the anomaly ratios are tiny: $p = 1\%, 2\%, 3\%, 4\%, 5\%$ (on text datasets we employ classic embeddings). Obviously seen from Fig. 2, our proposed SLA²P still attains favorable performance in comparison to other baseline approaches. Regarding all the text and image datasets mentioned above, SLA²P dominates in AUPR and AUROC with minimal numerical deviations. Furthermore, as demonstrated by the steadiness of the SLA²P curve in Fig. 2, SLA²P is remarkably stable with the alternation of p , especially in AUROC, despite the fact that we require the random sampling of the transformation matrices in our pipeline.

Results When the Normal Data are Multimodal: Many real-world applications for anomaly detection have a normal class that exhibits multimodal distribution [83], [84]. In the multimodal scenarios, all but one class is considered normal, yielding a dataset with a normal class that is multimodal and an abnormal class with fewer instances. We experiment on image and text datasets with anomaly ratio $p = 0.3$ and run 5 times under this more challenging setting. We, in turn, use one class of data

as anomalies and sample normal data from the rest categories. Other settings are the same as the previous unimodal setting (on text datasets we employ classic embeddings), with the mean outcomes being presented in Table VI. As shown in the table, our method performs the best consistently, and the performance improvement is even larger than the unimodal setting (over 30% AUROC gain on the Reuters dataset).

D. Sensitivity Analysis and Ablation Study

Impact of Adversarial Perturbation: To additionally showcase the role of the adversarial perturbations on the transformed features, we graph the distributions of anomaly scores prior to and following the perturbations. We choose experiments with normal sample class 'cat' of CIFAR-10, normal sample class 'non-insect invertebrates' of CIFAR-100, normal sample class 'comp.sys.mac.hardware' of 20 Newsgroups, and normal sample class 'topics' of Reuters. Additionally, we include the Arrhythmia experiment. As illustrated in Fig. 3, our self-supervised SLA framework has been capable of separating outliers: the anomaly scores of inliers are generally higher than those of outliers, which means the transformed features of inliers can be better classified by C_θ . However, outliers continue to exist, with their anomaly scores intertwining with those of inliers. However, following the perturbation, the scores for both regular and anomalous data decrease, but the inliers demonstrate greater resilience to the disturbances, making the anomalies more separable. This phenomenon shows that with the same magnitude of perturbation, the adversarial perturbation makes the transformed

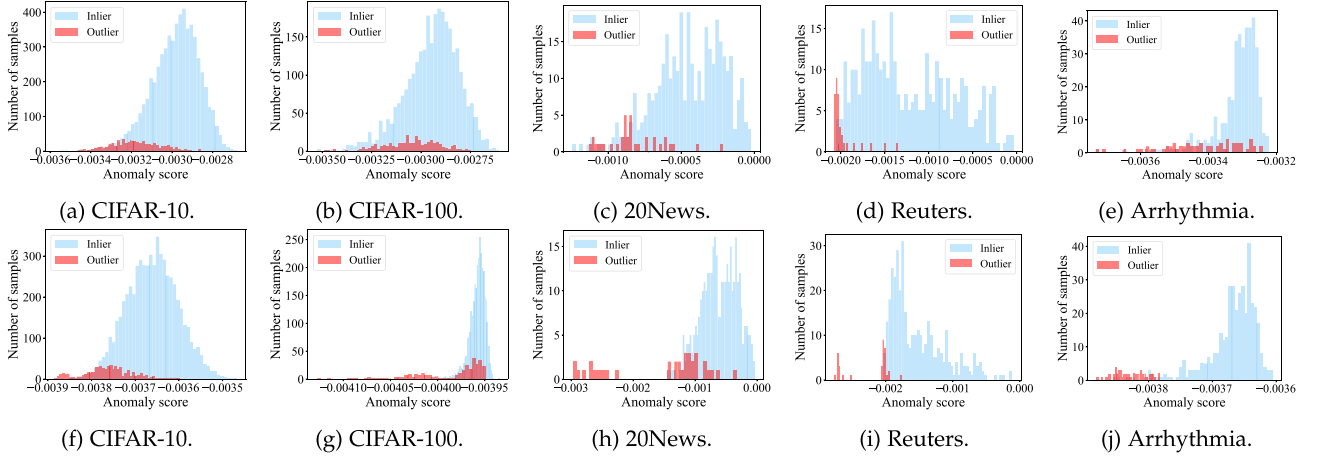


Fig. 3. Distribution visualization of the anomaly scores before (top row) and after (bottom row) adversarial perturbation of our framework.

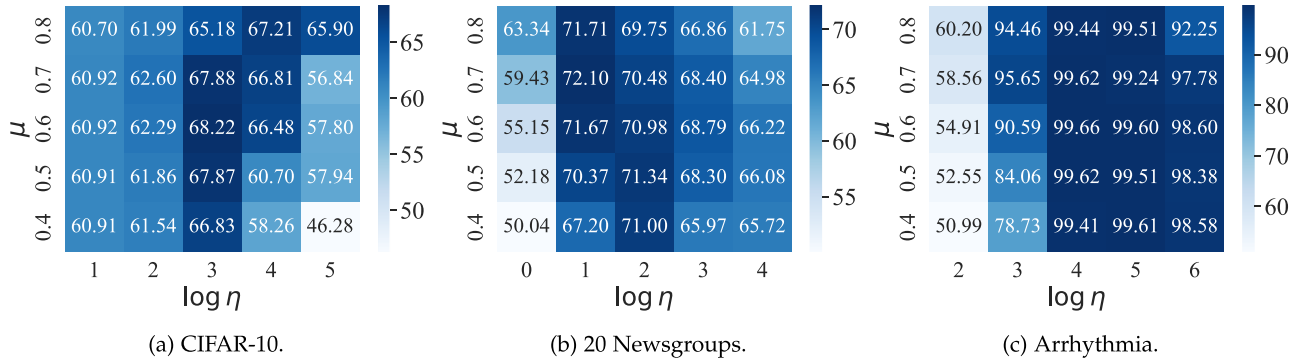


Fig. 4. AUPR heatmap of our SLA²P method.

features of the outliers have comparatively larger classification result degradation than those of the inliers in our self-defined pretext classification task.

Relationship Between μ and η : SLA²P exhibits high robustness to variations in the early stopping threshold μ and the perturbation magnitude η as demonstrated by the fine-grained AUPR score results we provide for CIFAR-10, 20 Newsgroups, and Arrhythmia datasets when $p = 0.1$, and μ is tuned to values in the range of 0.4, 0.5, 0.6, 0.7, 0.8 and η is tuned using a logarithmic scale, as shown in Fig. 4. We observed an extensive spectrum of hyperparameter combinations that yielded AUPR scores above 60% for CIFAR-10, over 55% for 20 Newsgroups, and over 50% for Arrhythmia, surpassing current state-of-the-art baselines. Moreover, within the span of hyperparameter options that exhibited competitive performance, we found that higher values of μ generally require larger values of η to ensure satisfactory detection performance. This makes sense as the more transformed samples are classified well, the larger the step size of the perturbations should be to separate the anomalies out.

Robustness to Transformed Dimension k : We investigate the robustness of AD performance of SLA²P to the transformation matrix dimension k . We perform an ablation analysis on Caltech 101 dataset with varying k in the interval [25, 350]. The experiment is repeated 5 times, and the AUROC and AUPR curve w.r.t. dimension k are depicted in Fig. 5. We can conclude that

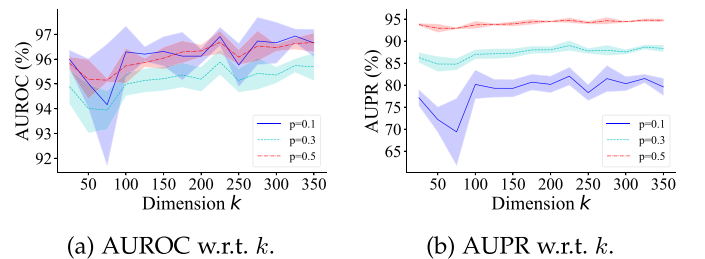


Fig. 5. UAD Performance on Caltech 101 with varying k of our proposed SLA²P approach.

the UAD performance of SLA²P is generally robust w.r.t. k with moderate growth when k increases and the growth tends to flatten out when k becomes comparatively larger. This phenomenon may be elucidated by Theorem 1 as k needs to be large enough to preserve the similarities between the transformed features.

VII. CONCLUSION

In this article, we present a novel structure, SLA²P, designed for unsupervised anomaly identification. Our method employs random projections as a new self-supervised technique at the feature level and combines it with adversarial perturbations to create distinctive anomaly scores. We present both theoretical

and empirical support for the efficacy of SLA²P. It is our hope that this research will serve as a foundation for using pretext task-based perturbation methods in other unsupervised learning scenarios or objectives.

REFERENCES

- [1] G. Pang, C. Shen, L. Cao, and A. V. D. Hengel, "Deep learning for anomaly detection: A review," 2020, *arXiv: 2007.02500*.
- [2] S. Min, B. Lee, and S. Yoon, "Deep learning in bioinformatics," *Brief. Bioinf.*, vol. 18, pp. 851–869, 2017.
- [3] S. Khan and T. Yairi, "A review on the application of deep learning in system health management," *Mech. Syst. Signal Process.*, vol. 107, pp. 241–265, 2018.
- [4] A. O. Adewumi and A. A. Akinyelu, "A survey of machine-learning and nature-inspired based credit card fraud detection techniques," *Int. J. Syst. Assurance Eng. Manage.*, vol. 8, pp. 937–953, 2017.
- [5] A. Shen, R. Tong, and Y. Deng, "Application of classification models on credit card fraud detection," in *Proc. IEEE Int. Conf. Serv. Syst. Serv. Manage.*, 2007, pp. 1–4.
- [6] K. Fu, D. Cheng, Y. Tu, and L. Zhang, "Credit card fraud detection using convolutional neural networks," in *Proc. Int. Conf. Neural Inf. Process.*, 2016, pp. 483–490.
- [7] S. C. Tan, K. M. Ting, and F. T. Liu, "Fast anomaly detection for streaming data," in *Proc. Int. Joint Conf. Artif. Intell.*, 2011, pp. 1511–1516.
- [8] D. Kwon, H. Kim, J. Kim, S. C. Suh, I. Kim, and K. J. Kim, "A survey of deep learning-based network anomaly detection," *Cluster Comput.*, vol. 22, pp. 949–961, 2019.
- [9] Z. Chen, Y. Tian, W. Zeng, and T. Huang, "Detecting abnormal behaviors in surveillance videos based on fuzzy clustering and multiple auto-encoders," in *Proc. IEEE Int. Conf. Multimedia Expo*, 2015, pp. 1–6.
- [10] W. Sultani, C. Chen, and M. Shah, "Real-world anomaly detection in surveillance videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 6479–6488.
- [11] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, pp. 1–58, 2009.
- [12] S. Liang, Y. Li, and R. Srikanth, "Enhancing the reliability of out-of-distribution image detection in neural networks," in *Proc. Int. Conf. Learn. Representations*, 2018, pp. 1–9.
- [13] D. Hendrycks, M. Mazeika, and T. G. Dietterich, "Deep anomaly detection with outlier exposure," in *Proc. Int. Conf. Learn. Representations*, 2019, pp. 1–9.
- [14] Y.-C. Hsu, Y. Shen, H. Jin, and Z. Kira, "Generalized odin: Detecting out-of-distribution image without learning from out-of-distribution data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 10951–10960.
- [15] W. Liu, X. Wang, J. D. Owens, and Y. Li, "Energy-based out-of-distribution detection," 2020, *arXiv: 2010.03759*.
- [16] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," 2019, *arXiv: 1901.03407*.
- [17] V. Chandola, A. Banerjee, and V. Kumar, "Outlier detection: A survey," *ACM Comput. Surv.*, vol. 14, 2007, Art. no. 15.
- [18] G. Somepalli, Y. Wu, Y. Balaji, B. Vinzamuri, and S. Feizi, "Unsupervised anomaly detection with adversarial mirrored autoencoders," 2020, *arXiv: 2003.10713*.
- [19] G. Kwon, M. Prabhushankar, D. Temel, and G. AlRegib, "Backpropagated gradient representations for anomaly detection," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 206–226.
- [20] S. Venkataramanan, K.-C. Peng, R. V. Singh, and A. Mahalanobis, "Attention guided anomaly detection and localization in images," 2019, *arXiv: 1911.08616*.
- [21] R. T. Schirmmeister, Y. Zhou, T. Ball, and D. Zhang, "Understanding anomaly detection with deep invertible networks through hierarchies of distributions and features," 2020, *arXiv: 2006.10848*.
- [22] Y. Wang, C. Qin, R. Wei, Y. Xu, Y. Bai, and Y. Fu, "Self-supervision meets adversarial perturbation: A novel framework for anomaly detection," in *Proc. Conf. Inf. Knowl. Manage.*, 2022, pp. 4555–4559.
- [23] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2000, pp. 93–104.
- [24] X. Yang, L. J. Latecki, and D. Pokrajac, "Outlier detection with globally optimal exemplar-based GMM," in *Proc. SIAM Int. Conf. Data Mining*, 2009, pp. 145–154.
- [25] J. Kim and C. D. Scott, "Robust kernel density estimation," *J. Mach. Learn. Res.*, vol. 13, pp. 2529–2565, 2012.
- [26] M. Ester et al., "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. Int. Conf. Knowl. Discov. Data Mining*, 1996, pp. 226–231.
- [27] Z. He, X. Xu, and S. Deng, "Discovering cluster-based local outliers," *Pattern Recognit. Lett.*, vol. 24, pp. 1641–1650, 2003.
- [28] M.-L. Shyu, S.-C. Chen, K. Sarinnapakorn, and L. Chang, "A novel anomaly detection scheme based on principal component classifier," Miami Univ Coral Gables FL Dept of Electrical and Computer Engineering, Tech. Rep., 2003.
- [29] R. Paffenroth, K. Kay, and L. Servi, "Robust PCA for anomaly detection in cyber networks," 2018, *arXiv: 1801.01571*.
- [30] B. Schölkopf, R. C. Williamson, A. J. Smola, J. Shawe-Taylor, and J. C. Platt, "Support vector method for novelty detection," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2000, pp. 582–588.
- [31] M. Amer, M. Goldstein, and S. Abdennadher, "Enhancing one-class support vector machines for unsupervised anomaly detection," in *Proc. ACM SIGKDD Workshop Outlier Detection Description*, 2013, pp. 8–15.
- [32] H. Zenati, C. S. Foo, B. Lecouat, G. Manek, and V. R. Chandrasekhar, "Efficient GAN-based anomaly detection," 2018, *arXiv: 1802.06222*.
- [33] P. Perera, R. Nallapati, and B. Xiang, "OCGAN: One-class novelty detection using GANs with constrained latent representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 2898–2906.
- [34] C.-H. Lai, D. Zou, and G. Lerman, "Novelty detection via robust variational autoencoding," 2020, *arXiv: 2006.05534*.
- [35] J. Chen, S. Sathe, C. C. Aggarwal, and D. S. Turaga, "Outlier detection with autoencoder ensembles," in *Proc. SIAM Int. Conf. Data Mining*, 2017, pp. 90–98.
- [36] S. Pidhorskyi, R. Almohsen, and G. Doretto, "Generative probabilistic novelty detection with adversarial autoencoders," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 6823–6834.
- [37] D. Abati, A. Porrello, S. Calderara, and R. Cucchiara, "Latent space autoregression for novelty detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 481–490.
- [38] B. Zong et al., "Deep autoencoding gaussian mixture model for unsupervised anomaly detection," in *Proc. Int. Conf. Learn. Representations*, 2018, pp. 1–9.
- [39] C. Lai, D. Zou, and G. Lerman, "Robust subspace recovery layer for unsupervised anomaly detection," in *Proc. Int. Conf. Learn. Representations*, 2020, pp. 1–10.
- [40] G. Lerman and T. Maunu, "An overview of robust subspace recovery," in *Proc. IEEE*, vol. 106, no. 8, pp. 1380–1410, Aug. 2018.
- [41] S. Wang et al., "Effective end-to-end unsupervised outlier detection via inlier priority of discriminative network," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2019, pp. 5962–5975.
- [42] M. Sabokrou, M. Fayyaz, M. Fathy, Z. Moayed, and R. Klette, "Deep-anomaly: Fully convolutional neural network for fast anomaly detection in crowded scenes," *Comput. Vis. Image Understanding*, vol. 172, pp. 88–97, 2018.
- [43] O. Rippel, P. Mertens, and D. Merhof, "Modeling the distribution of normal data in pre-trained deep features for anomaly detection," 2020, *arXiv: 2005.14140*.
- [44] L. Ruff et al., "A unifying review of deep and shallow anomaly detection," 2020, *arXiv: 2009.11732*.
- [45] J. Andrews, T. Tanay, E. J. Morton, and L. D. Griffin, "Transfer representation-learning for anomaly detection," *J. Mach. Learn. Res.*, 2016.
- [46] A. Kumagai, T. Iwata, and Y. Fujiwara, "Transfer anomaly detection by inferring latent domain representations," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2019, pp. 2471–2481.
- [47] V. Vercauteren, W. Meert, and J. Davis, "Transfer learning for anomaly detection through localized and unsupervised instance selection," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2020, pp. 6054–6061.
- [48] P. Burlina, N. Joshi, and I. Wang, "Where's wally now? Deep generative and discriminative embeddings for novelty detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 11507–11516.
- [49] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Representations*, 2015, pp. 1–8.
- [50] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD - A comprehensive real-world dataset for unsupervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 11507–11516.
- [51] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "Uninformed students: Student-teacher anomaly detection with discriminative latent

- embeddings,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 4183–4192.
- [52] L. Bergman, N. Cohen, and Y. Hoshen, “Deep nearest neighbor anomaly detection,” 2020, *arXiv: 2002.10445*.
- [53] T. S. Nazare, R. F. de Mello, and M. A. Ponti, “Are pre-trained CNNs good feature extractors for anomaly detection in surveillance videos?,” 2018, *arXiv: 1811.08495*.
- [54] G. Pang, C. Yan, C. Shen, A. van den Hengel, and X. Bai, “Self-trained deep ordinal regression for end-to-end video anomaly detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 12173–12182.
- [55] P. Bojanowski and A. Joulin, “Unsupervised learning by predicting noise,” in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 517–526.
- [56] S. Gidaris, P. Singh, and N. Komodakis, “Unsupervised representation learning by predicting image rotations,” in *Proc. Int. Conf. Learn. Representations*, 2018, pp. 1–12.
- [57] A. v. d. Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” 2018, *arXiv: 1807.03748*.
- [58] T. Chen, S. Kornblith, M. Norouzi, and G. E. Hinton, “A simple framework for contrastive learning of visual representations,” in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
- [59] I. Golan and R. El-Yaniv, “Deep anomaly detection using geometric transformations,” in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 9781–9791.
- [60] L. Bergman and Y. Hoshen, “Classification-based anomaly detection for general data,” in *Proc. Int. Conf. Learn. Representations*, 2020, pp. 1–9.
- [61] J. Tack, S. Mo, J. Jeong, and J. Shin, “CSI: Novelty detection via contrastive learning on distributionally shifted instances,” in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 11839–11852.
- [62] V. Schwag, M. Chiang, and P. Mittal, “SSD: A unified framework for self-supervised outlier detection,” in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 1–9.
- [63] I. J. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and harnessing adversarial examples,” in *Proc. Int. Conf. Learn. Representations*, 2014, pp. 1–9.
- [64] K. Lee, K. Lee, H. Lee, and J. Shin, “A simple unified framework for detecting out-of-distribution samples and adversarial attacks,” in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 7167–7177.
- [65] X. Gu, J. Sun, and Z. Xu, “Spherical space domain adaptation with robust pseudo-label loss,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9101–9110.
- [66] Z. Feng, C. Xu, and D. Tao, “Self-supervised representation learning by rotation feature decoupling,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 10 364–10 374.
- [67] G. W. Brier et al., “Verification of forecasts expressed in terms of probability,” *Monthly Weather Rev.*, vol. 78, pp. 1–3, 1950.
- [68] S. Dasgupta and A. Gupta, “An elementary proof of a theorem of Johnson and Lindenstrauss,” *Random Struct. Algorithms*, vol. 22, no. 1, pp. 60–65, 2003.
- [69] A. Krizhevsky et al., “Learning multiple layers of features from tiny images,” Master’s thesis, Dept. Comput. Sci., Univ. Toronto, 2009.
- [70] L. Fei-Fei, R. Fergus, and P. Perona, “Learning generative visual models from few training examples: An incremental Bayesian approach tested on 101 object categories,” in *Proc. Conf. Comput. Vis. Pattern Recognit. Workshop*, 2004, pp. 178–178.
- [71] K. Lang, “Newsweeder: Learning to filter netnews,” in *Machine Learning Proceedings*. San Mateo, CA, USA: Morgan Kaufmann, 1995.
- [72] D. Dua and C. Graff, “UCI machine learning repository,” 2017. [Online]. Available: <http://archive.ics.uci.edu/ml>
- [73] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” 2018, *arXiv: 1810.04805*.
- [74] T. Brown et al., “Language models are few-shot learners,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 1877–1901.
- [75] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proc. 8th IEEE Int. Conf. Data Mining*, 2008, pp. 413–422.
- [76] T. Reiss, N. Cohen, L. Bergman, and Y. Hoshen, “PANDA: Adapting pretrained features for anomaly detection and segmentation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 2805–2813.
- [77] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [78] A. Rajaraman and J. D. Ullman, *Mining of Massive Datasets*. Cambridge, U.K.: Cambridge Univ. Press, 2011.
- [79] S. Ioffe and C. Szegedy, “Batch normalization: Accelerating deep network training by reducing internal covariate shift,” in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.
- [80] B. Xu, N. Wang, T. Chen, and M. Li, “Empirical evaluation of rectified activations in convolutional network,” 2015, *arXiv:1505.00853*.
- [81] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. Int. Conf. Learn. Representations*, 2015, pp. 1–10.
- [82] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [83] F. Ahmed and A. Courville, “Detecting semantic anomalies,” in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2020, pp. 3154–3162.
- [84] M. J. Cohen and S. Avidan, “Transformally—two (feature spaces) are better than one,” 2021, *arXiv:2112.04185*.



Yizhou Wang (Member, IEEE) received the BS degree in mathematics and applied mathematics (Honors Program) from the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an, China, in 2020. He is currently working toward the PhD degree with the Department of Electrical and Computer Engineering, Northeastern University, Boston, Massachusetts, under the supervision of Prof. Yun Raymond Fu. His research interests include machine learning, computer vision and data mining. He has published some papers at top-tier conferences including ICLR, CVPR, CIKM, ICDM and IJCAI. He has served as a reviewer for *ACM Transactions on Knowledge Discovery from Data*, *Knowledge and Information Systems*, *ICML*, *NeurIPS*, *ICLR*, *CVPR*, *ECCV*, *KDD*, *AAAI*, *IJCAI*, *PAKDD*, *ICME*, etc.



Can Qin received the BE degree from Xidian University, China in 2018 and the PhD degree in computer engineering from Northeastern University, Boston, Massachusetts in 2023. He is now a research scientist with Salesforce Research. His research focus spans transfer learning, semi-supervised learning, and deep learning in broad. He received awards for the Best Paper Award of the Real-World Recognition from Low-Quality Images and Videos workshop at ICCV, 2019. Additionally, he has published some papers at top-tier conferences, including *NeurIPS*, *ICLR*, *AAAI*, *ICCV*, *ECCV*, *KDD*, etc.



Rongzhe Wei received the BS degree from Xi'an Jiaotong University in 2021. He is currently working toward the PhD degree with the Georgia Institute of Technology. His research interests are theoretical foundations of graph neural networks, and data privacy (especially for graph data).

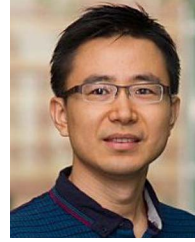


Yi Xu (Graduate Student Member, IEEE) received the BS and MS degrees from Xi'an Jiaotong University in 2017 and 2020 respectively. Currently he is working toward the PhD degree with Northeastern University. His current research interests are machine learning, computer vision, pattern recognition, and their applications to intelligent systems.



Systems, *IEEE Transactions on Knowledge and Data Engineering*, ICML, NeurIPS, CVPR, IJCAI, etc.

Yue Bai received the BSc degree in mathematics from Donghua University, Shanghai, China in 2017 and the MEng degree in data analytics engineering from Northeastern University, Boston, MA, in 2019. He is currently working toward the PhD degree with the Department of Electrical and Computer Engineering, Northeastern University, Boston, MA, USA. His research interests include machine learning, computer vision, and deep learning. He has served as the reviewer for *IEEE Transactions on Image Processing*, *IEEE Transactions on Neural Networks and Learning*



His research interests are machine learning, computational intelligence, Big Data mining, computer vision, pattern recognition, and cyber-physical systems. He has extensive publications in leading journals, books/book chapters, and international conferences/workshops. He serves as an associate editor, the chair, a PC member, and a reviewer of many top journals and international conferences/workshops. He received seven prestigious young investigator awards from NAE, ONR, ARO, IEEE, INNS, UIUC, and Grainger Foundation; eleven best paper awards from IEEE, ACM, IAPR, SPIE, and SIAM; and many major industrial research awards from Google, Samsung, Amazon, Konica Minolta, JP Morgan, Zebra, Adobe, and Mathworks. He is currently an associate editor of *IEEE Transactions on Pattern Analysis and Machine Intelligence*. He is member of Academia Europaea (MAE), member of European Academy of Sciences and Arts (EASA), fellow of National Academy of Inventors (NAI), AAAS Fellow, AIMBE fellow, IAPR fellow, OSA fellow, SPIE fellow, AAIA fellow, and ACM distinguished scientist.

Yun Fu (Fellow, IEEE) received the BEng degree in information engineering and the MEng degree in pattern recognition and intelligence systems from Xi'an Jiaotong University, China, and the MS degree in statistics and the PhD degree in electrical and computer engineering from the University of Illinois at Urbana-Champaign. He has been an interdisciplinary faculty member with the College of Engineering and the College of Computer and Information Science, Northeastern University, since 2012